

Design Decisions:

1. LIBRARY: Represents the main academic resource centers on campus.

Relationships:

- Each library is located in one building (BLDG_CODE).
- A library has many books.

2. BOOK: Represents books stored and managed by the library. Tracking their borrowing is essential.

Relationships:

- Each book belongs to one library.
- Books are borrowed by students via the BORROWS entity.

3. BORROWS: Captures the borrowing activity of books by students, including borrow and return dates.

Relationships:

- A student can borrow many books.
- Each borrow record is for one specific book.

4. STUDENT_CLUB: Represents student organizations and clubs. They have advisors and designated meeting rooms.

Relationships:

- Students are members of student clubs (many-to-many).
- Clubs have one advisor (professor).
- Clubs meet in rooms.

5. EVENT: Represents campus events such as workshops and social activities.

Relationships:

- Events are held in rooms.
- Events can be attended by students.
- Events can be organized by student clubs.

6. GYM: Represents athletic and fitness facilities available to students.

Relationships:

- Gyms are located in buildings.
- Students can use the gyms.